

Presented by Samira Hadid | 01/13

E-COMMERCE PRODUCT ENTITY-RESOLUTION & MATCHING PLATFORM

DATA STRUCTURES AND ALGORITHMS – 3
(25CS2103E)

BY:

Chakrika Panakala — 2520090042

Hrushika Habeeb — 2520090054

Section: 12

Guide: Ms. K. Chandusha
Assistant Professor, CS-IT

ABSTRACT

- E-commerce platforms contain multiple listings that may represent the same product.
- Listings can differ in names, descriptions, attributes, identifiers, and images.
- Our system uses a hybrid entity-resolution approach to identify matching products.
- It combines rule-based matching, fuzzy text matching, semantic similarity, and image similarity.
- Blocking and candidate generation reduce unnecessary comparisons and improve efficiency.
- The system produces unified product IDs and confidence-scored matches.
- Low-confidence matches can be sent for human review.





OBJECTIVES

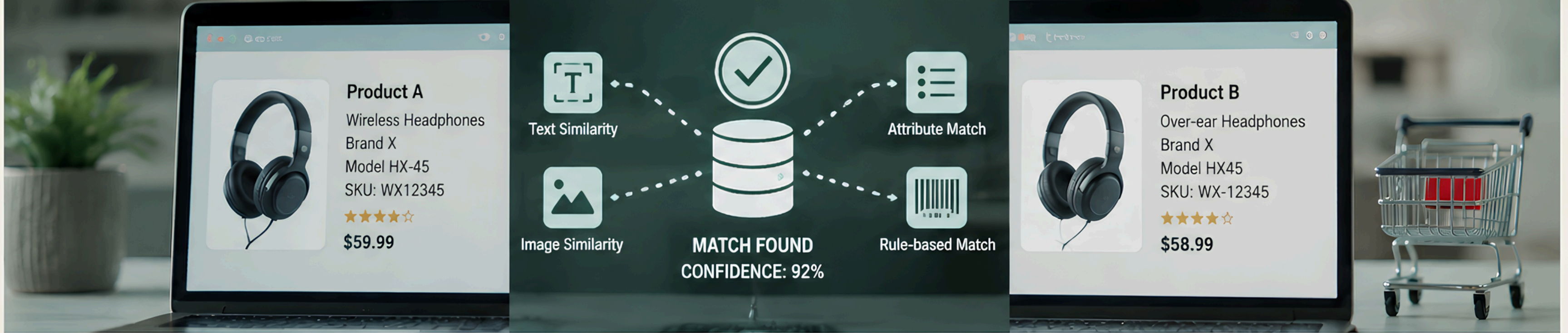
- Identify duplicate or equivalent products across different e-commerce listings.
- Generate a unified product ID for the same underlying product.
- Combine exact, fuzzy, semantic, and visual matching techniques.
- Reduce unnecessary pairwise comparisons using blocking and candidate generation.
- Differentiate between product variants such as size, color, storage, and model.
- Assign a confidence score to every potential match.
- Send low-confidence matches for human review.
- Improve catalog quality for price comparison and marketplace integration.

EXISTING SYSTEM

We focus on businesses that want to scale up in the digital era.

- ➞ Most e-commerce platforms depend heavily on SKU, UPC, GTIN, or product IDs.
- ➞ Different sellers may use different names and descriptions for the same product.
- ➞ Traditional exact matching fails when:
 - ➞ Product names are different
 - ➞ Attributes are missing
 - ➞ Spelling varies
 - ➞ Product identifiers are unavailable
- ➞ Manual catalog deduplication is time-consuming and difficult to scale.
- ➞ Simple string matching may incorrectly merge similar but different products.





PROPOSED SOLUTION

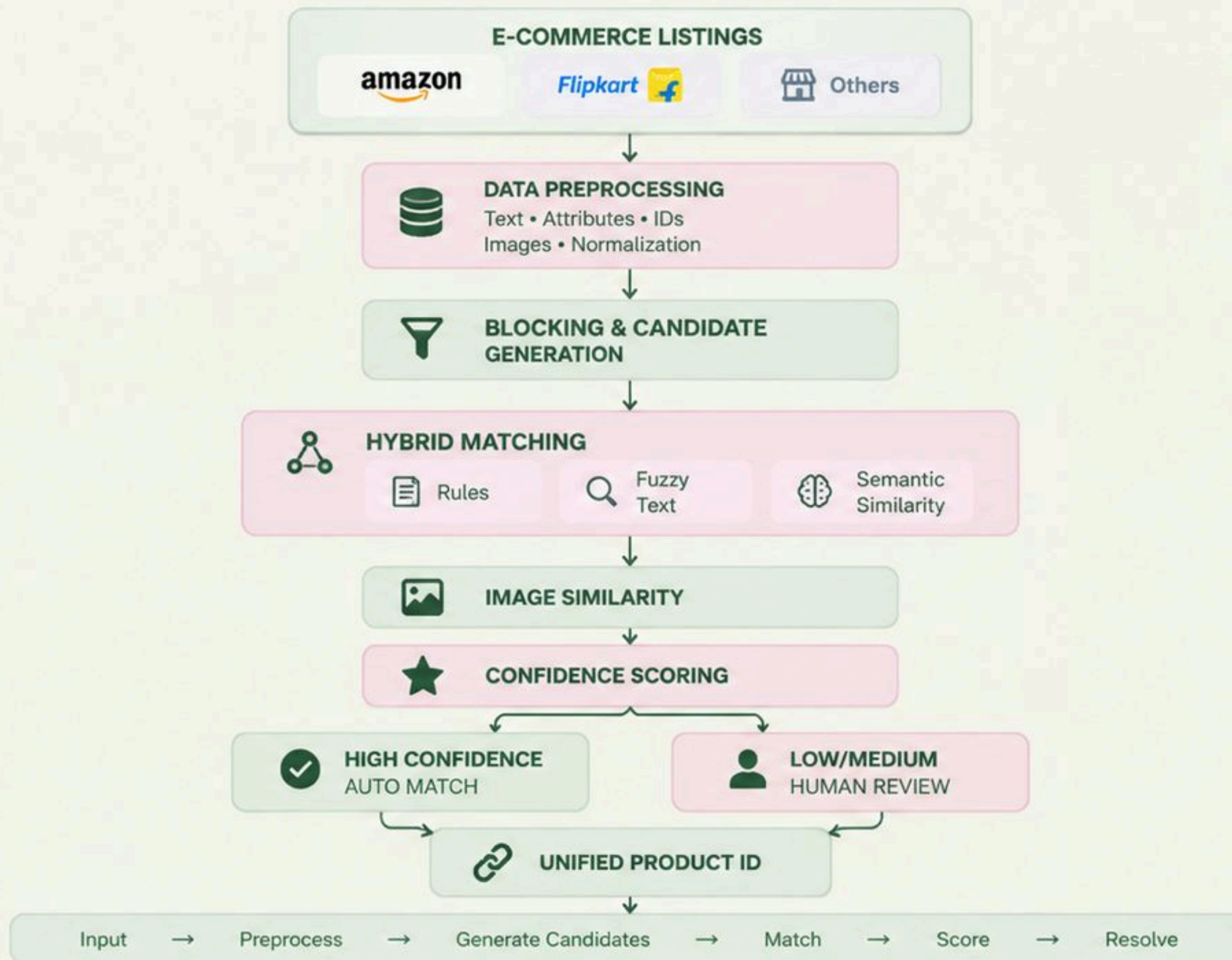
OUR APPROACH:

A hybrid entity-resolution system that combines multiple matching techniques to identify and group duplicate product listings.

KEY COMPONENTS:

- Rule-based matching using SKUs, UPCs, brands, and models
- Fuzzy text matching for product titles and descriptions
- Semantic similarity using embeddings to capture meaning
- Image-based similarity to compare product visuals
- Confidence scoring to classify matches (Match / Possible / Non-Match)
- Human-in-the-loop review for low-confidence matches
- Unified product ID generation for matched listings

SYSTEM ARCHITECTURE



INTEGRATION OF DSA CONCEPTS



ARRAYS / LISTS

→ Store and process product listings and attributes



HASHING

→ Fast lookup of SKU, UPC, model and product IDs

Aa

STRING ALGORITHMS

→ Fuzzy matching of product names and descriptions



SORTING

→ Rank candidates by similarity/confidence score



GRAPHS

→ Represent products as nodes and matching relationships as edges



Product Listings



DSA Processing

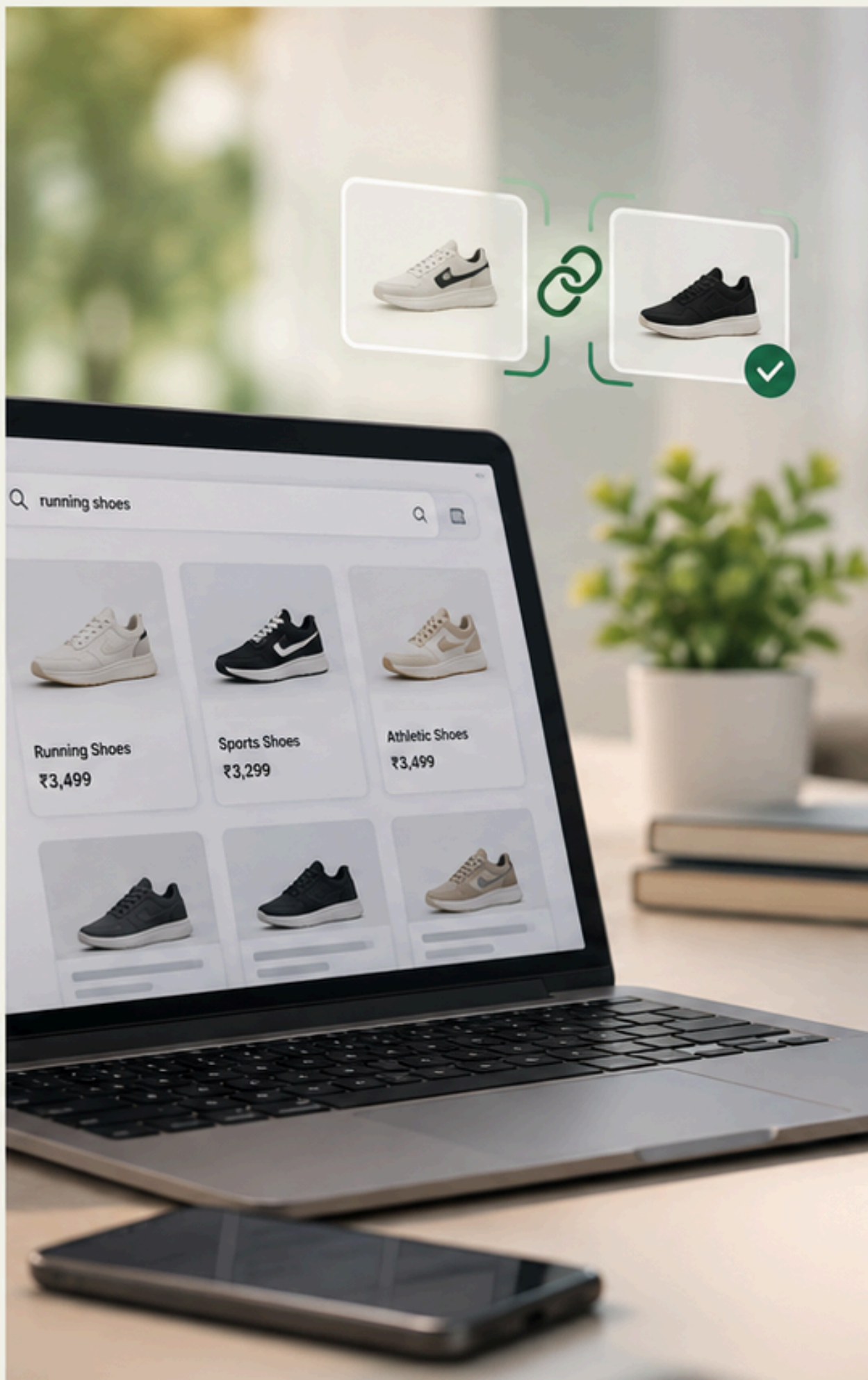


Candidate Matches



Unified Product ID





ADVANTAGES



IMPROVED MATCHING ACCURACY

Combines rule-based, fuzzy, semantic and image-based matching for more reliable product identification.



EFFICIENT PROCESSING

Blocking and candidate generation reduce unnecessary comparisons and improve scalability.



HANDLES DIVERSE LISTINGS

Works even when product names, descriptions, attributes, identifiers or images differ.



UNIFIED PRODUCT IDENTIFICATION

Generates a common product ID for listings representing the same product.



CONFIDENCE-BASED RESULTS

Assigns confidence scores to matches, making results easier to evaluate.



HUMAN-IN-THE-LOOP REVIEW

Low-confidence matches can be flagged for human verification, reducing incorrect matches.

CONCLUSION

- Hybrid approach effectively identifies matching products across multiple e-commerce platforms.
- Improves accuracy, efficiency and scalability through smart matching and blocking.
- Provides unified product IDs with confidence scores, with human review for uncertain matches.



THANK YOU!!